

Function Application Examples

$[Item, Dog, PersonRes, XRes]$

This file demonstrates function application in Z notation definitions.

Example 1 : Basic function definitions

$squareRes : \mathbb{N} \rightarrow \mathbb{N}$ $double : \mathbb{N} \rightarrow \mathbb{N}$
$\forall n : \mathbb{N} \bullet squareRes(n) = n * n$ $\forall n : \mathbb{N} \bullet double(n) = 2 * n$

Example 2 : Functions with multiple arguments

$add : \mathbb{N} \times \mathbb{N} \rightarrow \mathbb{N}$ $maximum : \mathbb{N} \times \mathbb{N} \rightarrow \mathbb{N}$
$\forall a, b : \mathbb{N} \bullet add(a, b) = a + b$ $\forall a, b : \mathbb{N} \bullet (maximum(a, b) = a \vee maximum(a, b) = b) \wedge maximum(a, b) \geq a \wedge maximum(a, b) \geq b$

Example 3 : Generic functions

$[XRes]$ $collection : seq\ XRes$

Example 4 : Function application elem quantifiers

$posFunc : \mathbb{N} \rightarrow \mathbb{N}$
$\forall x : \mathbb{N} \bullet posFunc(x) > 0 \Rightarrow posFunc(x) > 0$

Example 5 : Schema with function type

$Heights$ $height : PersonRes \rightarrow \mathbb{N}$
$\forall p : PersonRes \bullet height(p) > 0 \wedge height(p) < 300$

Example 6 : Nested function application

$func1 : \mathbb{N} \rightarrow \mathbb{N}$ $func2 : \mathbb{N} \rightarrow \mathbb{N}$ $func3 : \mathbb{N} \rightarrow \mathbb{N}$
$\forall x : \mathbb{N} \bullet func1(func2(func3(x))) \geq 0$